

## Idea I am proposing:

Predictive Food Safety Intelligence Platform that turns public inspection data into explainable early warning signals for consumers, public health agencies, platforms(Square, DoorDash, etc.) and food safety operators.

**Your Name: Jun Xu**

## Your background and skills that will benefit this project

I have 10+ years of analytics/data infra experience in fintech and operations. With strengths in SQL, Python, data pipelines, and deploying data products. Experience building and deploying consumer facing web apps.

## Problem

Foodborne illness is a major public health issue. The CDC estimates that 48 million Americans get sick from foodborne illness each year, with 128,000 hospitalizations and 3,000 deaths. ([Source](#))

Restaurant inspection data is public, but it is hard for everyday people to interpret. A consumer can look up past inspection results, but most people cannot easily answer a question like “Is this restaurant becoming riskier over time?”.

Chicago has shown that data science can improve inspection prioritization. The city previously used predictive analytics to forecast restaurants likely to have critical violations. ([Source](#))

**My proposal** is to build a public facing and regulator friendly Food Safety Intelligence platform that predicts future inspection risk and explains the key drivers behind the risk. Chicago as the MVP then scale food safety risk intelligence layer for cities, restaurant groups, insurers, and platforms.

## Impact

This project could benefit:

- **Consumers** - by making food safety information easier to understand.
- **Public health agencies / inspectors** - by helping prioritize inspection resources.
- **Restaurants** - by identifying operational risk signals before they become serious violations.
- **Platforms / insurers**
  - **Platforms** like Square, Toast, DoorDash and OpenTable
  - **Insurers** like Traverlars as a potential commercial extension.

## Market Size

There is a real market around food safety, compliance, and restaurant operations.

The global food safety testing market was estimated at \$26.27B in 2025 and projected to reach \$48.01B by 2033. ([Source](#))

The restaurant food safety compliance software market is estimated around \$1.2B–\$1.4B in 2024, depending on the market research source, with forecasts around \$3B+ by 2033.

The capstone would start with Chicago, but the broader opportunity is a scalable food safety risk intelligence layer for cities, restaurant groups, insurers, and compliance platforms.

## Market Landscape / Competitive Landscape / Existing companies solving the same / similar problem

Who are the major players and main vendors in the space? What are the existing solutions?

| Company Name                               | Stage (startup, enterprise)  | Product / Solution overview                                   | Who is the primary customer?               | Key differentiation vs your proposal                                                           |
|--------------------------------------------|------------------------------|---------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------|
| Chicago Food Inspection Forecasting        | Government / civic analytics | Predicts likely critical violations to prioritize inspections | Chicago Department of Public Health        | Strong precedent, but not a modern public-facing product or startup platform.                  |
| <a href="#">Safe LA Dining</a>             | Consumer software            | Restaurant health inspection score                            | Consumers                                  | Pure consumer facing. No API.                                                                  |
| Square / Toast / <a href="#">Owner.com</a> | Restaurant SaaS              | Help restaurants increase revenue                             | Restaurants                                | Mainly serves businesses to increase revenue                                                   |
| <a href="#">SafetyCulture</a>              | Enterprise SaaS              | Digital audits, checklists, compliance workflows              | Restaurants, hospitality, operations teams | Focuses on internal compliance execution, not citywide predictive risk from public data.       |
| <a href="#">Inspect2GO</a>                 | Government software          | Inspection, permitting, code enforcement workflows            | Local government agencies                  | Helps manage inspection operations, but not primarily consumer-facing predictive risk scoring. |

Existing tools are mostly either government workflow systems or internal compliance tools. This project focuses on predictive, explainable, public facing food safety intelligence built from open data.

## Relevant readings, market research, white papers, academic research

- CDC Foodborne Illness Facts and Stats - establishes public health importance. ([source](#))
- Chicago Food Inspection Forecasting - precedent for predictive inspection prioritization. ([source](#))
- Hindsight Analysis of the Chicago Food Inspection Forecasting Model - independent analysis of Chicago's prior model ([source](#))
- GovTech coverage of Chicago predictive inspection model - reports that predictive analytics helped identify critical violations earlier. ([source](#))

## Datasets

- Chicago Food Inspections - inspection results from restaurants from 2010 to present ([data source](#))
- Chicago 311 complaints - neighborhood level signals such as sanitation, garbage, rodents, and related complaints ([311 service request](#))
- Chicago Business Licenses - can help identify business status, license type, and restaurant metadata ([Current active licenses](#), [historical business licenses](#))
- NOAA weather data - optional signal for temperature and extreme weather etc ([NOAA Climate Data Online API](#))
- Yelp Open Dataset - optional signal for review trends ([Yelp Open Dataset](#))

## Possible data science techniques and why?

- Binary Classification - predict probability of failed inspection or critical violation
- Feature engineering - prior violations, days since last inspection, repeat violation patterns
- Geospatial modeling - nearby complaint density, neighborhood risk trends.
- NLP - parse violation descriptions into categories
- Explainability - SHAP or feature attribution to explain why a restaurant is high risk
- Model evaluation - AUC, precision, recall

## Final Deliverables

- Data pipeline for Chicago inspection + complaint + license data
- Model predicting critical violation / failed inspection risk
- Explainable risk scores
- Interactive Chicago map
- Restaurant detail page
- API endpoint for risk scoring
- Short business case for commercialization